

- 6 The role of sensory receptor cells in mammals is to detect stimuli and generate action potentials in sensory neurones.

Human taste buds on the tongue contain chemoreceptor cells. Different chemoreceptor cells respond to different chemical stimuli.

Fig. 6.1 is a diagram of chemoreceptor cells in a taste bud.

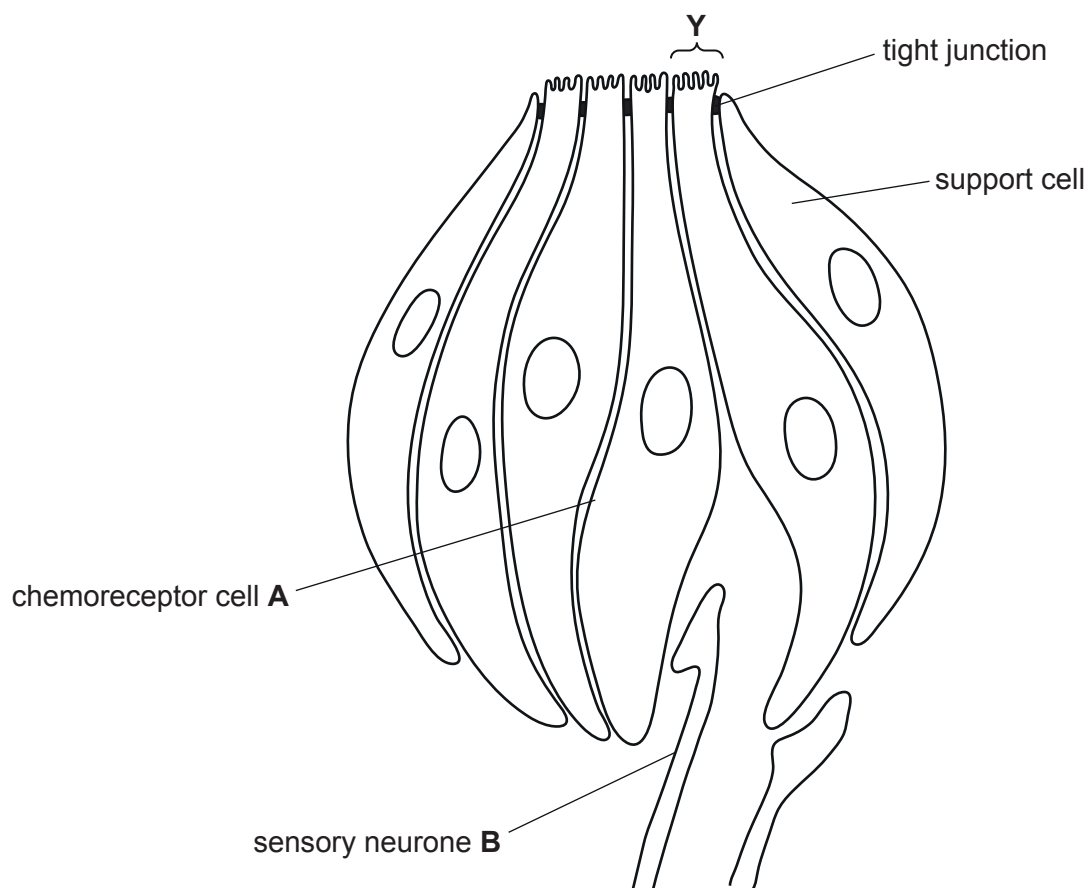


Fig. 6.1

- (a) Name the structures in the region **Y** and describe their function in a chemoreceptor cell.

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..... [3]

- (b)** When a sugar molecule binds to a receptor protein on the cell surface membrane of cell **A**, calcium ions are released into the cytoplasm of the cell by the endoplasmic reticulum.

Explain how the release of calcium ions will lead to an action potential being generated in sensory neurone **B**.

[6]

[Total: 9]